FOP MINI PROJECT:
ROCK, PAPER AND SCISSOR GAME

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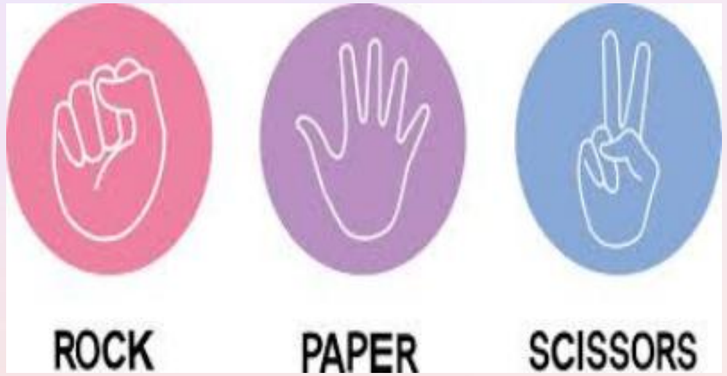
KEY FEATURES OF OUR GAME:

- **INTERACTIVE USER INPUT VALIDATION:**
  uses printf() and scanf() for interaction and validates user input to prevent invalid choice.

- **EFFICIENT USE OF MODULO OPERATOR:**
  rand() %3 ensures the random number stays within valid game options(0, 1, 2).

RANDOM COMPUTER CHOICE GENERATION:

uses rand() and srand() to generate unpredictable computer moves.



STRUCTURED GAME LOGIC USING CONDITION:

implements game rules using if-else or switch case for accurate decision making.

MULTIPLE ROUNDS WITH SCORE TRACKING:

uses loops to allow replay and variables to maintain user and computer scores.

UNDERSTANDING OF HOW OUR GAME FUNCTIONS:

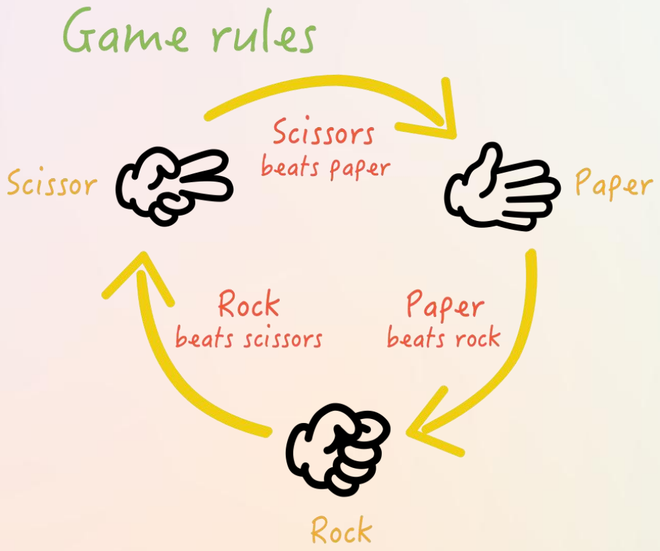
- The program starts in main() and initializes variables.
- User enters a choice (rock, paper or scissors).
- Computer generates a random choice using rand().
- Conditional statements compare both choices.
- The result (win/ lose/ tie) is displayed.
- The loop allows the game to continue till either the user or the computer earns 3 points and wins the game.

THE GAME LOGIC:

We will have to translate the hand gestures into numbers so the computer understands them, and then handle all the possible win/loss combinations and also makes it user friendly.

To make this work in C, we map the choices to integers-

- 0 = ROCK
- 1 = PAPER
- 2 = SCISSORS



C PROGRAM CODE:

```c
#include <stdio.h>
#include <stdlib.h>
#include <time.h>

int main()
{
int userChoice, compChoice;
int userScore = 0, compScore = 0;

srand(time(0));

printf("=== Welcome to the Ultimate RPS Championship ===\n");
printf("First to 3 points wins the match!\n\n");

while (userScore < 3 && compScore < 3)
{
printf("Current Score -> You: %d | Computer: %d\n", userScore,
compScore);
printf("Choose: (0) Rock, (1) Paper, (2) Scissors: ");
```

```c
if (scanf("%d", &userChoice) != 1 || userChoice<0 || userChoice > 2)
{
printf("Invalid input! Please enter 0, 1, or 2.\n\n");
while(getchar() != '\n');
continue;
}

compChoice = rand() % 3 ;

printf("You chose: ");
if(userChoice == 0) printf("Rock\n");
else if(userChoice == 1) printf("Paper\n");
else printf("Scissors\n");

printf("Computer chose: ");
switch(compChoice)
{
case 0: printf("Rock\n");
break;
case 1: printf("Paper\n");
break;
case 2: printf("Scissors\n");
break;
}
```

```
if (userChoice == compChoice)
{
printf(">> It's a tie!\n\n");
}
else if ((userChoice == 0 && compChoice == 2) || (userChoice == 1 &&
compChoice == 0) || (userChoice == 2 && compChoice == 1))
{
printf(">> You win this round!\n\n");
userScore++;
}
else
{
printf(">> Computer wins this round!\n\n");
compScore++;
}
}
```

```
printf("Final Score -> You: %d | Computer: %d\n", userScore,
compScore);
if (userScore == 3)
{
printf("CONGRATULATIONS! You are the champion.\n");
}
else
{
printf("GAME OVER. The machines have won this time.\n");
}

return 0;
}
```

OUTPUT:

```
=== Welcome to the Ultimate RPS Championship ===
First to 3 points wins the match!

Current Score -> You: 0 | Computer: 0
Choose: (0) Rock, (1) Paper, (2) Scissors: 0
You chose: Rock
Computer chose: Paper
>> Computer wins this round!

Current Score -> You: 0 | Computer: 1
Choose: (0) Rock, (1) Paper, (2) Scissors: 2
You chose: Scissors
Computer chose: Scissors
>> It's a tie!

Current Score -> You: 0 | Computer: 1
Choose: (0) Rock, (1) Paper, (2) Scissors: 1
You chose: Paper
Computer chose: Rock
>> You win this round!

Current Score -> You: 1 | Computer: 1
Choose: (0) Rock, (1) Paper, (2) Scissors: 2
You chose: Scissors
Computer chose: Rock
>> Computer wins this round!
```

```
Current Score -> You: 1 | Computer: 2
Choose: (0) Rock, (1) Paper, (2) Scissors: 1
You chose: Paper
Computer chose: Paper
>> It's a tie!

Current Score -> You: 1 | Computer: 2
Choose: (0) Rock, (1) Paper, (2) Scissors: 0
You chose: Rock
Computer chose: Paper
>> Computer wins this round!

Final Score -> You: 1 | Computer: 3
GAME OVER. The machines have won this time.


=== Code Execution Successful ===
```

WHAT WE LEARNED:

- Writing structured c programs
- Using conditional statements effectively
- Working with loops
- Generating random numbers
- Improving debugging skills
- Enhancing logical thinking

REAL LIFE APPLICATIONS:

- Used in decision making situations
- Basic concepts used in game development
- Helps understand AI-based decision logic
- Foundation for developing larger interactive programs

CONCLUSION:

- successfully developed a stone paper scissor game in c
- improved understanding of programming logic
- learned how to use random numbers and conditions

Thank You.
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